

MAT 426, Class 16: Orthogonal Diagonalization

Example 1. Find the QR factorization of $A = \begin{bmatrix} 1 & 1 \\ 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}$ using the Householder QR algorithm with four-digit arithmetic.

In Class 15, we showed that $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 0.0001 \\ 0 \end{bmatrix}$. Computing $A^{(1)} = H_1 A$ implicitly gave us

$$A^{(1)} = H_1 A = \begin{bmatrix} -1 & -1 \\ 0 & -0.0001 \\ 0 & 0.0001 \end{bmatrix}.$$

Full vs Reduced QR Factorization Suppose A is an $m \times n$ matrix.

- The QR factorization via Gram-Schmidt orthogonalization produces an orthogonal matrix Q of size _____ and upper triangular matrix R of size _____. This is called the _____ QR factorization of A .
- The QR factorization via Householder matrices produces an orthogonal matrix Q of size _____ and upper triangular matrix R of size _____. This is called the _____ QR factorization of A .
- How can the full QR factorization $A = QR$ be obtained from the reduced one $A = Q_1 R_1$?
 - (1) Add _____ columns to Q_1 (using _____ orthogonalization) so that it becomes an $m \times m$ orthogonal matrix Q
 - (2) Append _____ rows of _____ to R_1 so that it becomes an $m \times n$ upper triangular matrix R .

Orthogonal Diagonalization

DIAGONALIZATION THEOREM *An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .*

Example 2. If possible, find a matrix P that diagonalizes $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.

Example 3. If possible, find a matrix P that diagonalizes $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$.

Definition. A square matrix A is **orthogonally diagonalizable** if there exists an _____ matrix Q and a _____ D such that $Q^T A Q = ___$.

THEOREM *If A is orthogonally diagonalizable, then A is symmetric.*

Proof.

THEOREM *If A is a real symmetric matrix, then the eigenvalues of A are real.*

Proof.

THEOREM *If A is a symmetric matrix, then any two eigenvectors corresponding to distinct eigenvalues of A are orthogonal.*

Proof.

THEOREM *Let A be an $n \times n$ real matrix. Then A is symmetric if and only if it is orthogonally diagonalizable.*

Proof.

Example 4. Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.

SPECTRAL THEOREM *An $n \times n$ symmetric matrix A has the following properties:*

- (a) A has n real eigenvalues, counting multiplicities*
- (b) The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.*
- (c) The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal*
- (d) A is orthogonally diagonalizable.*